Break it down

Programming · Level 1 · Lesson 4



Today we will

- Name the four parts of computational thinking.
- Compare two algorithms by counting steps.
- Plan a computational-thinking moment in your subject.

Remember

- Decompose, find patterns, abstract, write the algorithm.
- Halving: 7 guesses for 1 to 100, about 10 for 1,000.
- Halving only works on things in order.

Guess my number

1. Warm-up (5 min): Guess my number, freely; count the guesses.
2. Teach (10 min): The four parts, with school examples.
3. Halving (10 min): Play again by halving; compare counts.
4. Plan (10 min): Find the four parts in a task from your subject.
5. Wrap-up (5 min): Which part was hardest to spot? Exit ticket.

Keep the board tally of guesses: the drop from the warm-up to halving is the lesson's best evidence.

Let’s discuss

1. Why does halving need the numbers in order?
2. Which part of computational thinking do you already teach without naming it?
3. Where else could we compare two ways of doing a job by counting steps?

Exit ticket

- Name the four parts, with one example of each from today.